02 Amazon Redshift Serverless Setup Guide

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Key Concepts

- **Work Groups:** Resource allocation units for compute
- **Namespaces:** Resource segregation for storage, users, and databases
- **RPUs:** Redshift Processing Units (each RPU = 16GB memory)

Prerequisites

- AWS Account with appropriate permissions
- Note: New accounts may be eligible for $300 in free credits

Step-by-Step Setup

1. Initial Setup

1. Navigate to Amazon Redshift console
2. Click **"Try Redshift serverless free trial"**
3. Click **"Try Redshift serverless"**
4. Under configurations, select **"Customize Settings"**

2. Namespace Configuration

1. **Name:** Leave as "default-namespace" (or customize)
2. **Admin Credentials:**
   - Check **"Customize Admin user credentials"**
   - Username: `admin` (or customize)
   - Password: Enter a secure password
   - ⚠️ **Important:** Store this password securely - you'll need it for pipeline connections

3. IAM Role Setup

1. Under **Permissions**, click dropdown
2. Select **"Create an IAM role"**
3. Choose **"Select any S3 bucket"**
4. Click **"Create IAM role"**
5. Check the box to select the created role
6. 📝 **Note:** Save the role name - you may need to add permissions later

4. Work Group Configuration

1. **Name:** Leave as "default-workgroup" (or customize)
2. **Capacity:** Select **8 RPUs** (minimum recommended)
   - Range available: 8-512 RPUs

- 8 RPUs = 128GB total memory
- Sufficient for course workloads

3. **VPC Settings**: Leave default VPC selected
4. **Security Group**: Leave default selected
5. **Subnets**: Ensure all default subnets are selected

5. Finalize Creation

1. Click **"Save Configuration"**
2. Wait 2-3 minutes for serverless environment creation
3. Click **"Continue"** when complete
4. Reload the page to see your namespace and work group

6. Query Editor Access

1. Click **"Query data"** (top right)
2. Access the **V2 Query Editor**
3. Expand **"default-workgroup"** to see:
   - Sample database (with dummy data)
   - `dev` database

Network Configuration (Critical for External Access)

7. Enable Public Access

1. Return to Redshift console
2. Click on your **work group** under **Data Access** tab
3. Under **Network & Security**, find **"Publicly accessible"** (disabled by default)
4. Click **"Edit"**
5. Check **"Turn on publicly accessible"**
6. Click **"Save Changes"**
7. Wait for update to complete

8. Configure Security Group

1. Under **Data Access** → **Network & Security**, click **Security Group**
2. Select the **Group ID**
3. Click **"Edit Inbound Rules"**
4. Click **"Add Rule"**
5. Configure the rule:
   - **Type**: Select "Redshift"
   - **Port**: Auto-populated to 5439
   - **Source**: `0.0.0.0/0` (allows connections from anywhere)
6. Click **"Save Rules"**

Important Notes

Security Considerations

- **Public Access:** Only recommended for development/learning
- **Production:** Use VPC endpoints and restrict IP ranges
- **Authentication:** Username/password still required for connections

Cost Management

- No charges for creating work groups/namespaces
- Charges apply only when executing queries
- Monitor usage in AWS Cost Explorer

Connection Details

- **Endpoint:** Available in Redshift console under work group details
- **Port:** 5439
- **Database:** `dev` (default)
- **Username:** `admin` (or your custom username)
- **Password:** Your configured password

Troubleshooting

Common Issues

1. **Connection Timeout:** Check security group rules
2. **Authentication Failed:** Verify username/password
3. **Public Access:** Ensure publicly accessible is enabled
4. **Subnet Issues:** Verify all subnets are selected

Next Steps

- Connect from external tools (Airflow, Python, etc.)
- Create additional databases and schemas
- Set up data pipelines
- Configure monitoring and alerting

Resources

- [AWS Redshift Serverless Documentation]
- [Redshift SQL Reference]
- [Best Practices Guide]

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